

Download the following file from the course web site: `tsd.m`

In this assignment you will design a digital state-feedback tracking system to control the position of the motor-driven cart used in Lab 1. The motor-driven system has the following state-space model, where **the first state variable is motor angular position and the second state variable is motor angular velocity**.

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ 0 & -\alpha \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ \beta \end{bmatrix} u.$$

Use the values of α and β from Lab 1. **The plant output is $y = x_1$.** [Note the Simulink block for the plant must be configured to send out both state variables for the state-feedback tracking system.]

1. Start with $T_s=0.3$. Use the guideline to choose the sampling interval for a digital state-feedback tracking system designed for step inputs.
2. **The additional dynamics model for tracking step inputs with a single-input plant is $(\Phi_a = 1, \Gamma_a = 1)$.** Choose spoles using only scaled Bessel poles. **Although the plant has an SDPP, we do not need to use it because the stability robustness bounds for the all-Bessel tracking system are very good.** The plant is second order and the additional dynamics are first order. Thus, the design model is third order and we choose $\text{spoles} = s3/T_s$.
3. Use `tsd.m` to calculate the tracking system gains and stability robustness bounds.
`[K1,K2,delta1,delta2]=tsd(phi,gamma,C,phia,gammaa,zpoles,T,'place')`
4. In Simulink, go to *Model Settings*, then *Solver*, change the Type to *Fixed-step* with a Fixed-step size of $T/100$.
5. Simulate the response to a **60-radian step command** using the block diagram at the bottom of this page and check the maximum plant input. The state-space model uses units of radians so the height of the step signal is simply 60. Decrease T_s until the maximum plant input is just under 5 volts.
6. What is the actual settling time and percent overshoot? The following commands will give you that information:

```
max_y=max(y(:,2))
PO=(max_y-60)/60*100 %percent overshoot
max_u=max(u(:,2))
ind=find(y(:,2)>0.99*60);
ST=y(min(ind),1) %settling time
```

